AI-NATIVE ENGINEERING MASTERY

Transform Your Development Practice

Trainer: Pasi Vuorio | ModernPath

YOUR TRAINER

PASI VUORIO

Founder & AI Architect

ModernPath

ModernPath
AI-Native Software Development

- 30+ years software engineering
- Enterprise AI transformation
- Brownfield modernization expert

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TRAINING OVERVIEW

SESSION 1

Fundamentals

- Build rapid prototypes with AI
- From zero to working demo

SESSION 2

Engineering

- Production-grade development
- Specs, TDD, architecture

SESSION 3

Scaling

- Optimize & scale
- Multi-agent, teams, metrics

Proto → Production → Scale

SESSION 1

AI-Assisted Development Fundamentals

90 min | Beginner

ModernPath

1. AI CODING TOOLS LANDSCAPE

Cursor

GitHub Copilot

Claude Code

Codex

RULES & CONFIGURATION

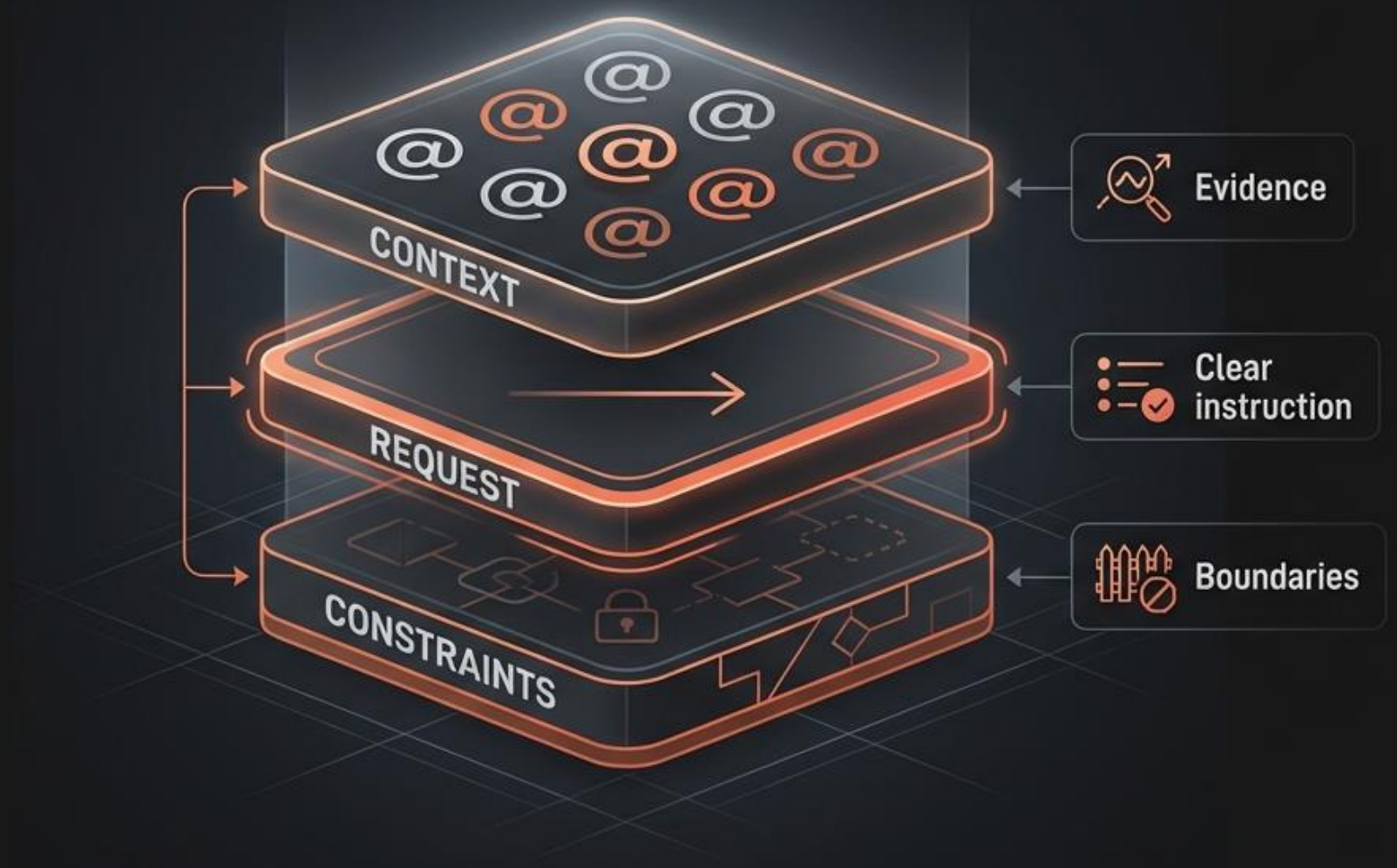
.cursor/rules/

CLAUDE.md

AGENTS.md

Your Codebase Constitution

2. EFFECTIVE PROMPTING PATTERNS



Company logo | © Copyright

3. MODEL SELECTION

CLAUDE

Reasoning: ★★★★☆
4/5 stars

Speed: ★★★★☆
4/5 stars

Context: 200K tokens

Best for:

- Best coding agent
- Nuanced writing
- Agentic workflows

GPT

Reasoning: ★★★★½
4.5/5 stars

Speed: ★★★☆☆
3/5 stars

Context: 128K tokens

Best for:

- High reasoning
- Solving difficult problems
- Debugging

GEMINI

Reasoning: ★★★★☆
4/5 stars

Speed: ★★★½☆
3.5/5 stars

Context: 1M+ tokens

Best for:

- Multimodal tasks
- Visual reasoning
- Frontend coding

Match the brain to the task

4. GREENFIELD vs BROWNFIELD

GREENFIELD

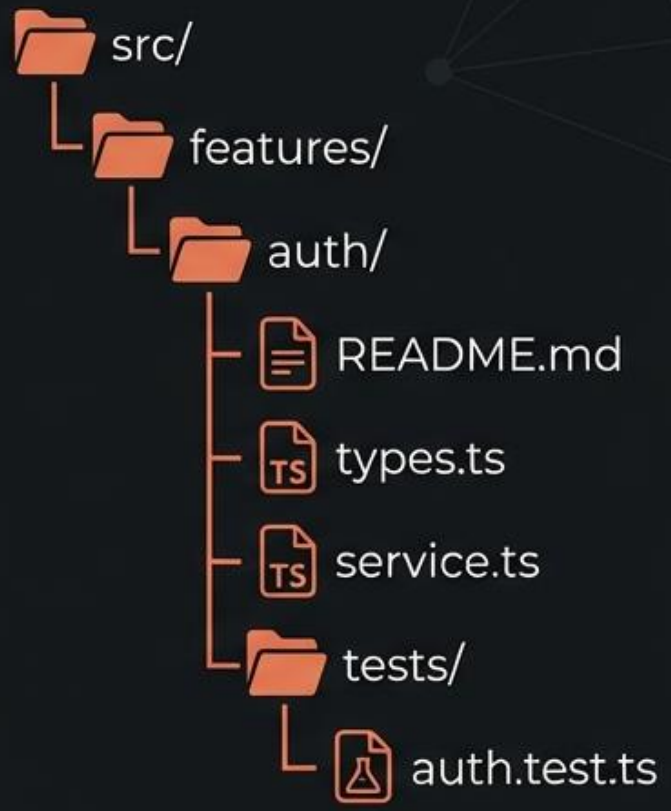
New Projects - Design for AI

BROWNFIELD

Existing Systems - Document First

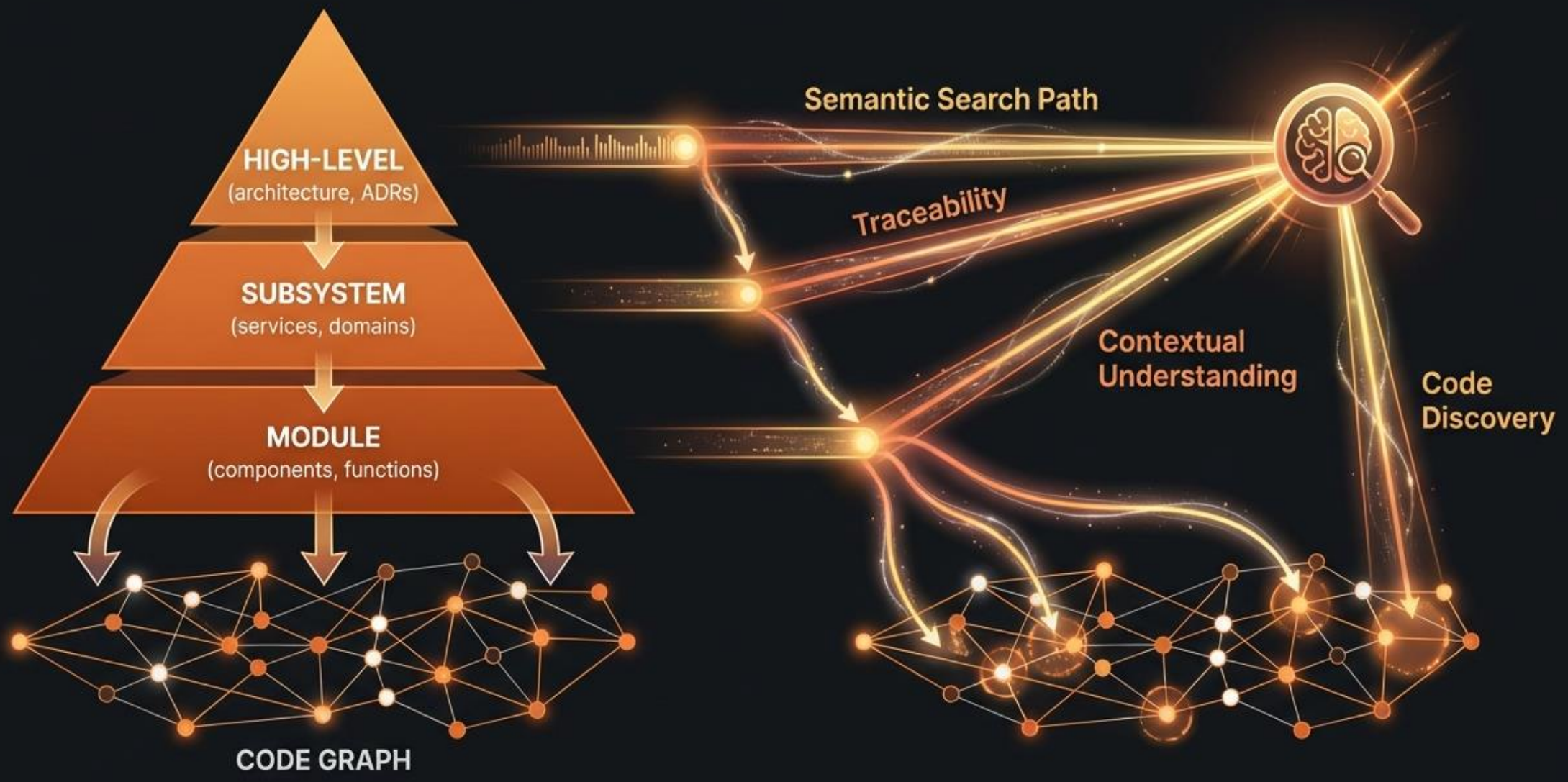
80% of enterprise work is brownfield

5. GREENFIELD: SPEC-DRIVEN DEVELOPMENT



Write the contract before the code

6. BROWNFIELD: 3-TIERED DOCUMENTATION



Your docs are the map for the AI brain

DOCUMENTATION PLAN: content-plan.md

GREENFIELD

- Plan before you build
- Define modules upfront
- AI follows your roadmap

BROWNFIELD

- Map what exists
- Index for AI navigation
- Progressive documentation

One index file. Both worlds.

SESSION 2

90 min | Intermediate

Spec-Driven Development & TDD with AI

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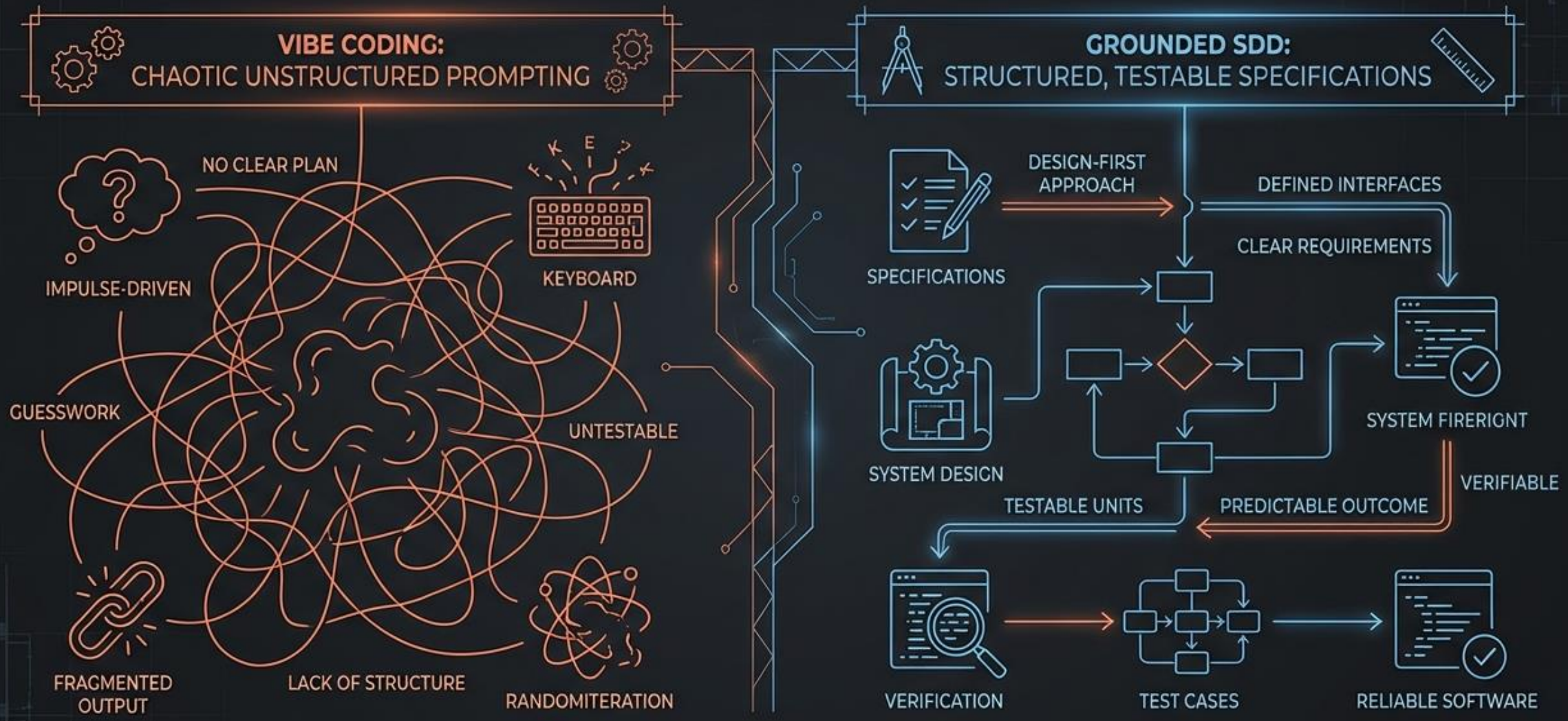
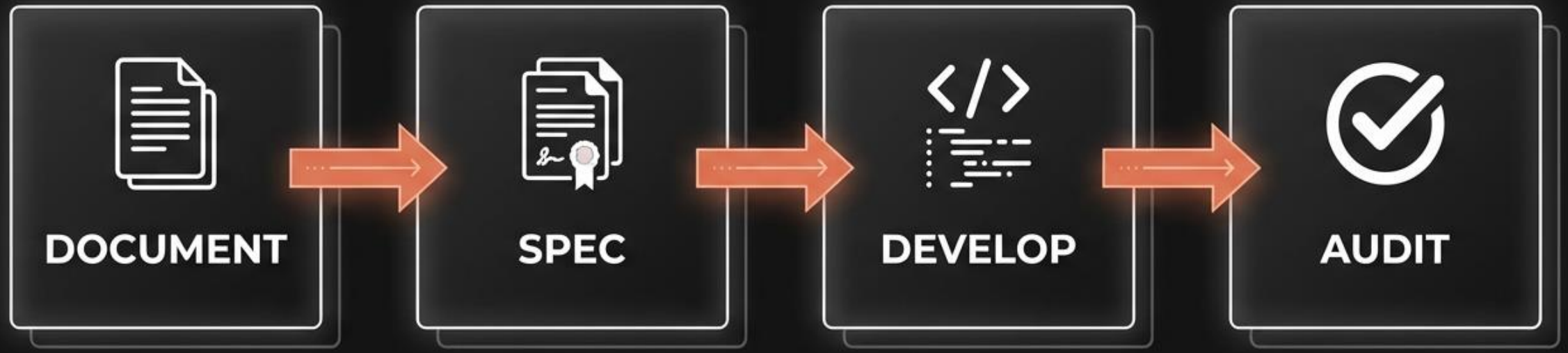
VIBE CODING VS. GROUNDED SDD



MODERNPATH 4-STEP PROCESS

The Complete AI Development Lifecycle



WRITING EFFECTIVE SPECIFICATIONS

SPEC DOCUMENT TEMPLATE

- OVERVIEW
- USER STORIES
- ACCEPTANCE CRITERIA
- TECHNICAL CONSTRAINTS
- TEST STRATEGY

GIVEN-WHEN-THEN FORMAT EXAMPLE

Given a registered user is on the login page

And the user enters valid credentials

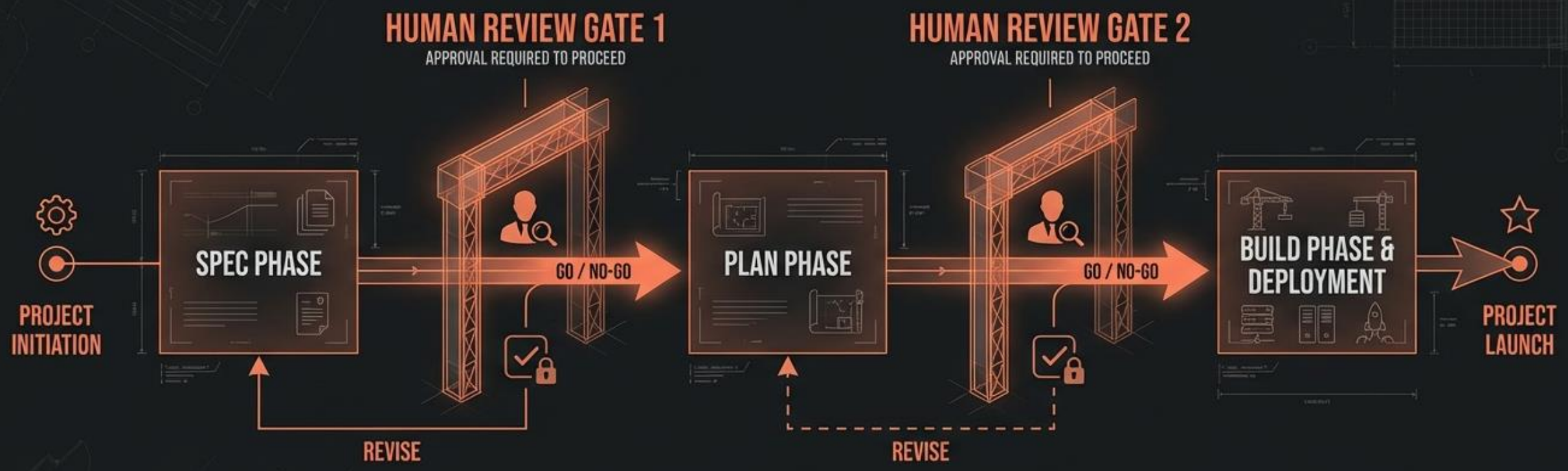
When the user clicks the "Login" button

Then the user is redirected to the dashboard

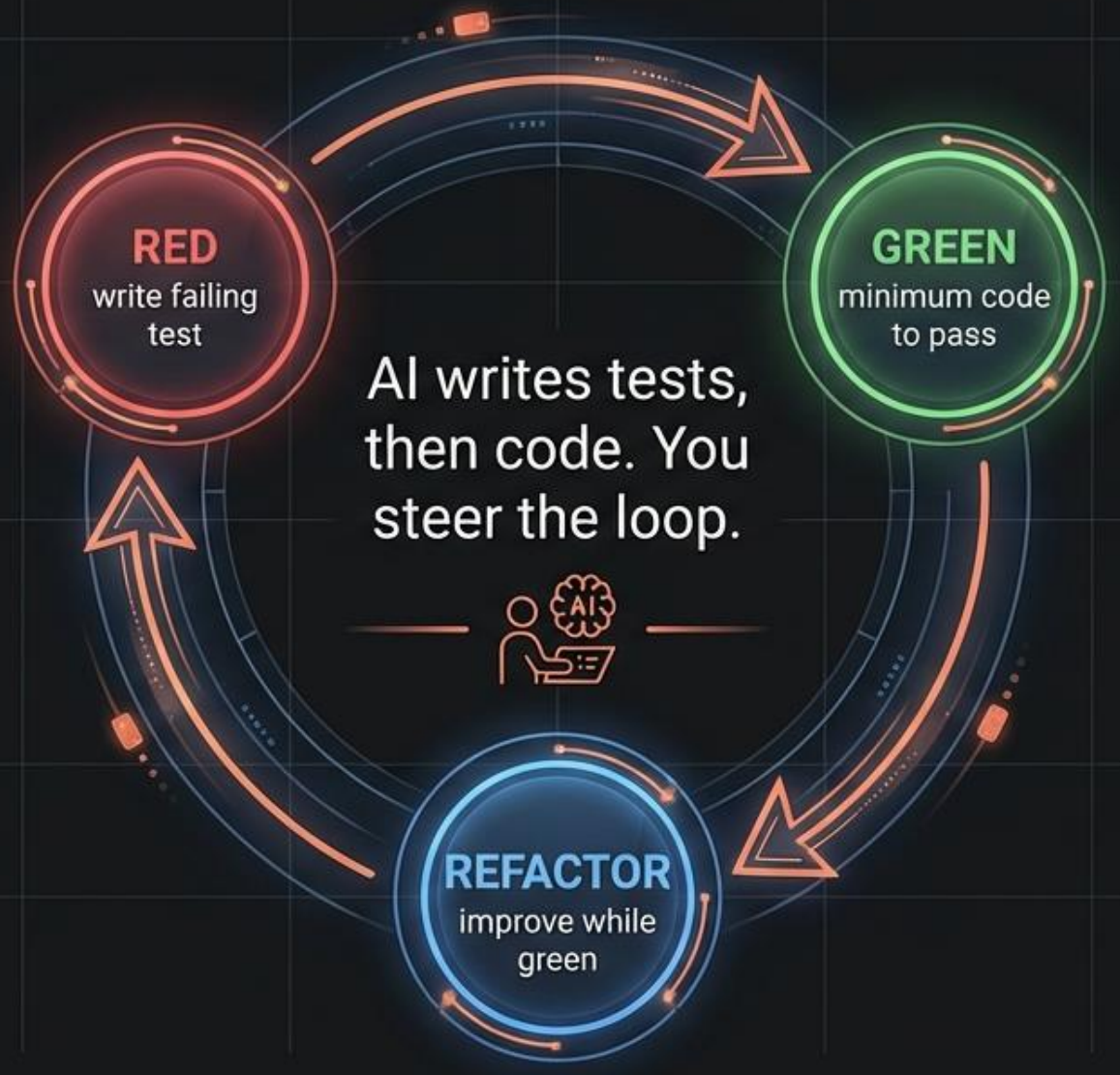
And a welcome message is displayed.

INSIGHT: IF AI CANNOT CONVERT YOUR AC TO A TEST, REWRITE IT

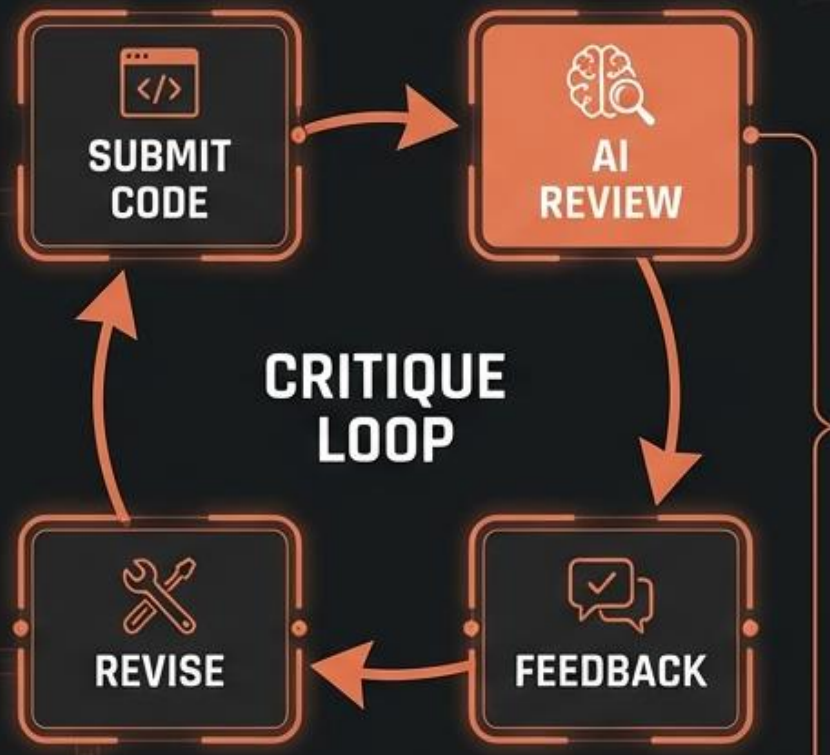
MANDATORY CHECKPOINTS



TDD CYCLE WITH AI



CODE REVIEW WITH AI



Continuous improvement cycle

AI CATCHES WHAT HUMANS MISS AT 3AM

Automated, tireless analysis for better code quality.

PR REVIEW PROMPT EXAMPLE

Review for security vulnerabilities

e.g., SQL injection, XSS

PR REVIEW PROMPT EXAMPLE

Check for edge cases

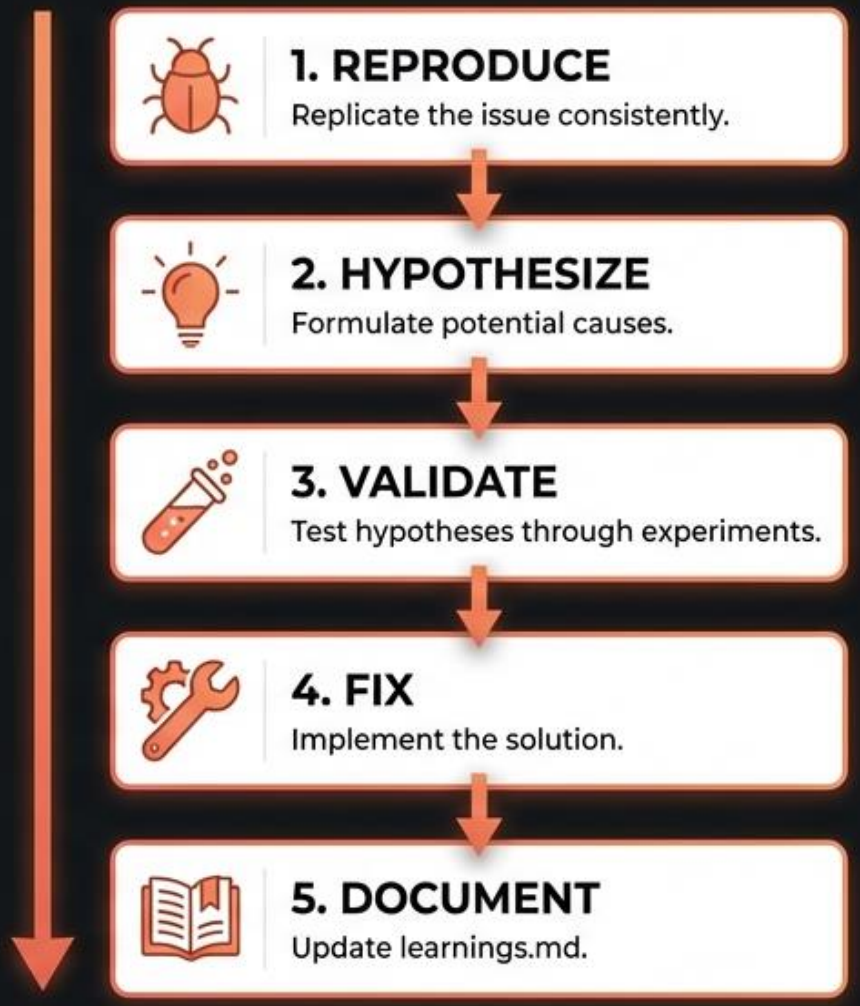
e.g., null values, boundary conditions

PR REVIEW PROMPT EXAMPLE

Suggest performance improvements

e.g., loop optimization, data structures

THE FIX WORKFLOW



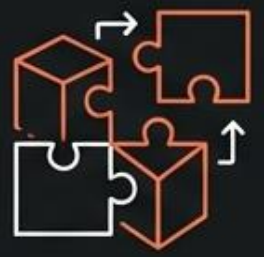
SIDE NOTE:
Max 3 attempts before asking user

EVERY FIX BECOMES TEAM KNOWLEDGE

AI-FRIENDLY ARCHITECTURE

MODULARITY

Decoupled components for independent development and scaling.



Decoupled components for independent development and scaling.

CLEAR CONTRACTS

Explicit interfaces and data schemas for reliable interaction.

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OBSERVABILITY

Built-in metrics, logging, and tracing for system health and insight.

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SESSION 3

90 min | Advanced

Advanced Agentic Workflows

3

AGENTIC ARCHITECTURE

MCP vs CLI TOOLS

CLI

PROS

- ✓ 30%+ better token efficiency
- ✓ Unix-style composability
- ✓ Human-readable debugging
- ✓ Mature ecosystem

CONS

- ✗ Can hallucinate syntax
- ✗ Harder for stateful tasks
- ✗ Platform-dependent

MCP

PROS

- ✓ Type-safe JSON schemas
- ✓ Complex integrations
- ✓ Human-in-loop approvals
- ✓ Growing ecosystem

CONS

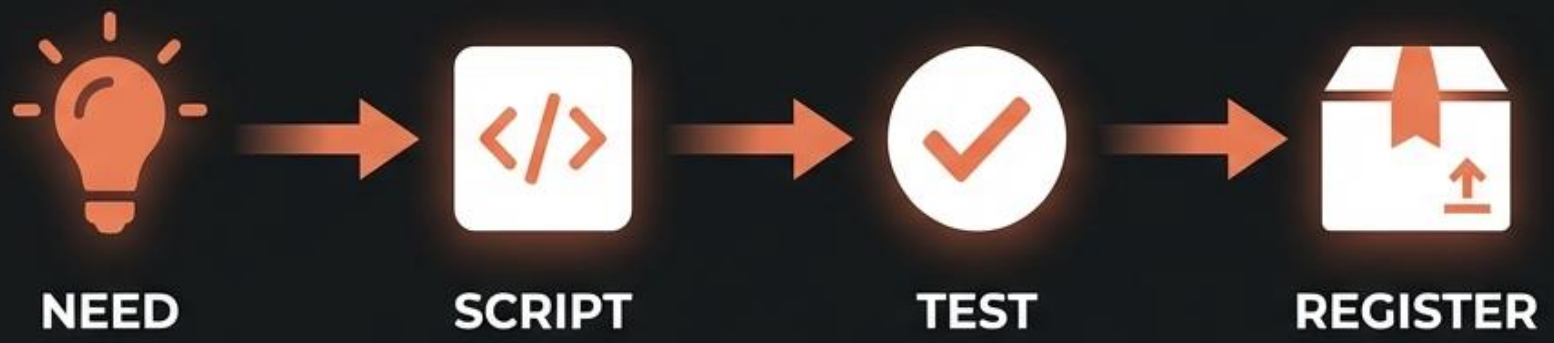
- ✗ Context bloat
- ✗ Operational complexity
- ✗ Often worse in benchmarks

CLI outperforms for existing tools. **MCP** for specialized/secure workflows.

MCP INTEGRATIONS

Connect AI to your company nervous system

INVENTING CLI TOOLS



EXAMPLE WORKFLOW

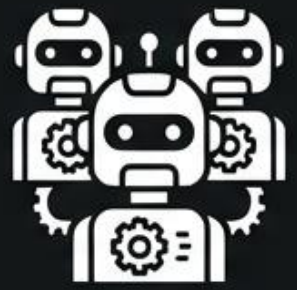
1. Identify repetitive task
2. Write TypeScript/Python tool
3. Add to package.json scripts
4. Document in AGENTS.md

TOOLS/ FOLDER STRUCTURE

```
tools/
├→ my-tool.ts
├─ data-processor.py
├─ README.md
├─ AGENTS.md
└─ package.json
```

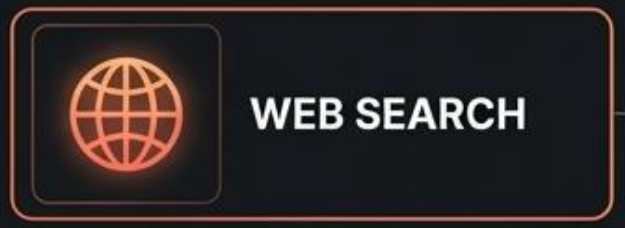
Your custom tools become AI superpowers

SKILLS vs SUB-AGENTS vs HOOKS

SKILLS	SUB-AGENTS	HOOKS
		
WHAT: Capabilities • Add abilities (search, APIs) • Example: Web search	**WHO: Specialists** • Delegate to experts • Example: Code reviewer	**WHEN: Control points** • Verify before/after • Example: Stop before commit

WHEN TO USE: Skills expand functionality, Sub-Agents handle complex tasks, and Hooks enforce rules at key stages.

AGENT SKILLS



WEB SEARCH

FILE OPERATIONS

CODE ANALYSIS

EXTERNAL APIS

Extend AI capabilities with pluggable skills

CLAUDE SKILLS

```
{
  "name": "search_docs",
  "description": "Search
internal documentation",
  "parameters": {
    "query": "string"
  }
}
```

CURSOR SKILLS

```
{
  name: "analyze_codebase",
  type: "analysis",
  path: "/src/app"
}
```

CUSTOM SKILL DEFINITIONS

```
def custom_api_call(data):
  return
requests.post("https://api.example.com/data",
  json=data)
```

Skills = Superpowers for your agents

SPECIALIZED SUB-AGENTS

SPEC WRITER

Translates high-level requirements into detailed technical specifications, user stories, and acceptance criteria for the development team. Ensures clarity and completeness.

Focus: Requirements Engineering & Clarity

TEST GENERATOR

Automatically creates comprehensive test suites, including unit tests, integration tests, and end-to-end scenarios based on specifications. Enhances code coverage and quality.

Focus: Quality Assurance & Test Automation

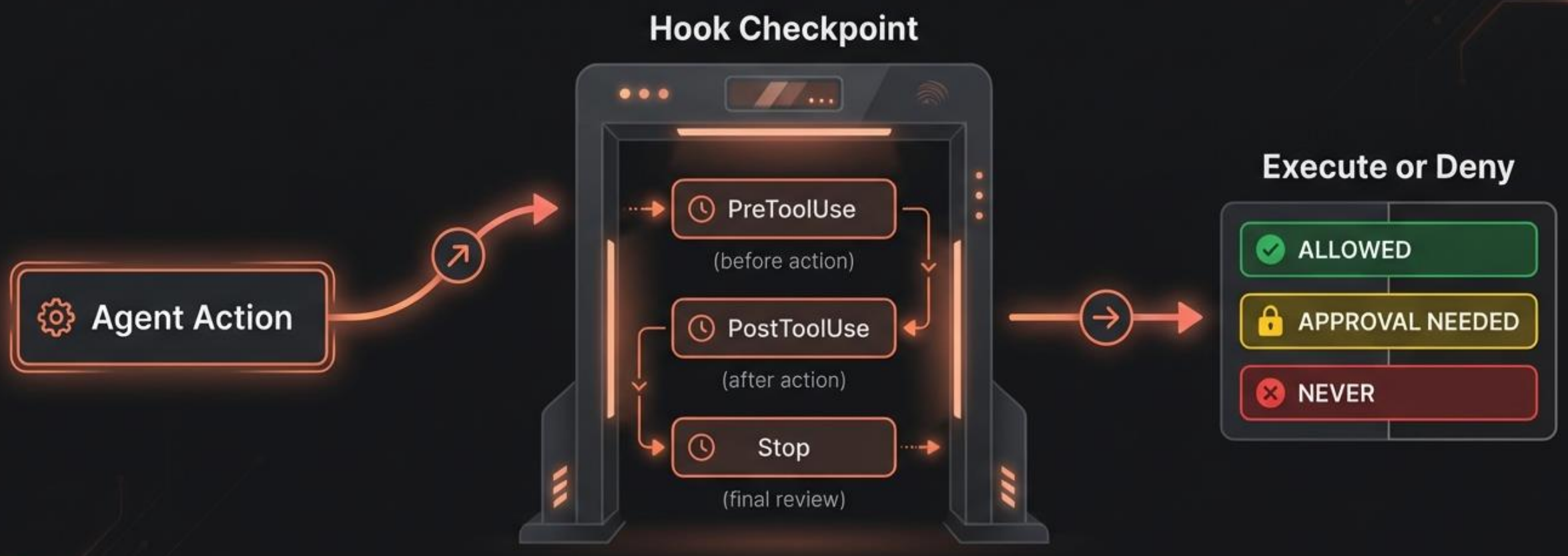
CODE REVIEWER

Analyzes code submissions for best practices, potential bugs, security vulnerabilities, and adherence to coding standards. Provides actionable feedback.

Focus: Code Quality & Security Compliance

One agent cannot do it all. Build a team of specialists.

HOOKS & VERIFICATION



Never let AI commit
without a green light

SCALING ACROSS TEAMS

ORGANIZATION
AGENTS.md - shared company standards

↓

PROJECT
Project-specific rules & context

↓

PERSONAL
Individual instructions & preferences

METRICS

- Time to PR
- AI assist ratio
- First-pass success

Centralize standards, localize specifics